Questions

Q1: #pragma omp parallel for 编译指令和 #pragma omp for 编译指令有何区别? 哪个效率更高?

#pragma omp for tells OpenMP to parallelize the for loop with existing team of threads

Q2：为何要对循环体进行调度（把iterations分配到不同线程）？有哪些调度类型?

schedule (type , chunksize)

Type can be:

- **static**: the iterations can be assigned to the threads before the loop is executed.
- **dynamic or guided**: the iterations are assigned to the threads while the loop is executing.
- **auto**: the compiler and/or the run-time system determine the schedule.
- **runtime**: the schedule is determined at run-time.

Q3: Dynamic schedule与Guided schedule有何区别?

- Each thread also executes a chunk, and when a thread finishes a chunk, it requests another one.
- However, in a guided schedule, as chunks are completed, the size of the new chunks decreases.

Q4: OpenMP提供了哪些机制来保障对于临界区访问的互斥?

- OpenMP provides several mechanisms for insuring mutual exclusion in critical sections.
  - Critical directives
  - Named critical directives
  - Atomic directives
  - Simple locks

Q5: #pragma omp critical(name) 与 #pragma omp critical 有何区别？用处在哪里?

two blocks protected with critical directives with different names can be executed simultaneously.